EvoState: Long-Horizon Evolving State & Inference Scaling

One-Page Scientific Concept & Evaluation Summary | Primary Sources: Sun et al. (2025), Gu & Dao (2023), Snell et al. (2024)

DataForge 2026 Pathway Track
Track: Foundation AI & Memory

Status: Live & Precomputed Verified

CENTRAL FALSIFIABLE SCIENTIFIC CLAIM:

"A fixed-size evolving state can carry useful information across sequences without storing every previous token, but increasing sequence length and conflicting updates can cause interference and information loss; additional inference-time computation can sometimes improve recovery."

1. Theoretical Foundations & Mechanics

- **KV-Cache vs Bounded State:** Autoregressive Transformers maintain $O(T \cdot d)$ KV buffers. Bounded memory maintains $O(1)$ state M_t via outer product binding: $M_t = \lambda M_{t-1} + \Delta_t (v_t \otimes k_t^T)$.
- **Superposition & Capacity Limits:** Under orthogonal keys, retrieval is exact: $\hat{v} = M_T q$. When $N > \alpha_c \cdot d$ (Hopfield/Associative limit $\alpha_c \approx 0.14$), cross-talk noise causes graceful superposition interference.
- **Inference-Time Scaling (BDH-CQ Probing):** Test-time refinement optimizes query sharpening without updating model weights: $v^{(k+1)} = (1-\eta)v^{(k)} + \eta M_T q_{sharp}^{(k)}$, reversing superposition degradation.

2. Architectural Taxonomy & Comparison

Architecture	State Complexity	Inference Latency	Interference Behavior
Transformer (KV)	$O(T \cdot d)$ Unbounded	$O(T)$ per step	Zero (Exact storage)
Fixed RNN (Mamba)	$O(d)$ Bounded	$O(1)$ Constant	Information loss over long lag
Associative (BDH)	$O(d_v \times d_k)$ Matrix	$O(1)$ Constant	Soft crosstalk superposition
BDH-CQ (Scaling)	$O(d_v \times d_k)$ Matrix	$O(K)$ Controllable	Active energy settling recovery

3. Empirical Sweep Results (2,700 Trials)

- **Delayed Recall (Lag $k \in [4, 128]$):** Educational Toy maintains **100% accuracy** across all lags via selective Δ_t gating, whereas standard vector RNN decays to 0%.
- **Sequence Clutter ($T \in [32, 512]$):** Full History maintains 100% at $O(T)$ latency penalty (7.4ms → 61.1ms). Toy maintains $O(1)$ latency with mild noise accumulation.
- **Destructive Overwrites ($N \in [1, 8]$):** Recency dominance observed; overwrite saturation drops accuracy from 100% ($N \leq 2$) to ~45% ($N \geq 5$).
- **Subspace Capacity Stress (N_{pairs} vs $d=32$):** Abrupt phase transition at $N > 4$ pairs, perfectly matching theoretical associative rank limits.
- **Test-Time Recovery:** Additional inference effort ($K=1$ to 16) yields measurable SNR gains when signal is in superposition, but cannot recover erased state.

4. Evidence Classification Standard

- **[PUBLISHED]** BDH $O(N)$ linear time & synapse duality (Sun et al., 2025).
- **[EMPIRICAL]** 2,700 seeded trials in data/precomputed/sweeps_master.csv.
- **[EDUCATIONAL TOY]** $d=32$ fast-weight surrogate for browser experimentation.
- **[NO FAKE COMPUTE]** Explicit badge distinction for Live vs Precomputed.

CRITICAL BOUNDARIES, LIMITATIONS & JUDGE DEFENSE:

1. Educational Toy vs Production BDH: Our browser toy is a pedagogical $d=32$ associative matrix surrogate. It does NOT claim to match production 10B BDH. | **2. Biological Brain Analogy:** Synapse/neuron mathematical duality is illustrative; no biological claims are made. | **3. Live vs Precomputed:** Live experiments execute in PyTorch/FastAPI/WASM; heavy sweeps are labelled Precomputed with full seed transparency. | **4. 60-Second Learner Journey:** Learner drives 4 controls (L, M, p, K) and evaluates the claim against raw unmanipulated ground truth.

Primary References: [1] Sun et al. (2025) *BDH: Beyond Transformer with Deep Associative Memory*. [2] Gu & Dao (2023) *Mamba*. [3] Snell et al. (2024) *Scaling LLM Test-Time Compute Optimally*. [4] Graves et al. (2014) *Neural Turing Machines*. — **Code & Data:** Open Source MIT / Deterministic Seeded (PyTorch 2.0+)